NICOLA **ZOMER**

M.Sc. student in Physics of Data (UniPD)

+39 3280569369

Padua, Italy

nicola.zomer@studenti.unipd.it

Personal Website

Linkedin

GitHub

About me

In October 2021 I started the master's degree in Physics of Data at the University of Padua, which aims at combining advanced knowledge in Physics with high-level training in Data Science.
I'm mainly interested in machine learning and data analysis techniques, combined with a statistical approach.

I strongly believe in teamwork, a skill I have developed having been part of a basketball team, a scout group for almost 10 years and having worked for the past two years in a university team (Alba CubeSat Unipd). In my spare time I also enjoy practicing handbalancing, a discipline that taught me patience, dealing with frustration, truly believing in the process and focus.

Education

Oct 2021 – Present

Master's degree in Physics of Data, *DFA "G. Galilei", University of Padua (IT).*

The Master's degree provides new theoretical and computational tools to tackle the explosion of datasets within the physicist mindset. The program intends to build a figure that combines advanced knowledge in the field of Physics with a high-level training in Data Science.

Aug 2022

Oxford Machine Learning Summer School 2022

The school covered some of the most important topics in ML/DL (e.g., statistical/probabilistic ML, representation learning, reinforcement learning, causal inference, vision & NLP, geometrical DL, ...). It consisted of 2 tracks: ML x Health and ML x Finance, plus a bonus module on ML Foundamentals.

Aug 2022

Qiskit Global Summer School 2022

- Description: Completed the two-week intensive course provided by IBM Quantum, completing all graded lab work assignments with a final cumulative score of 100%.
- Certificate of Quantum Excellence

2018 – 2021

Bachelor's degree in Aerospace Engineering, *University of Padua (IT).*

- Thesis title: *"A study of the Levi-Civita regularization of the gravity field"*
- Final grade: **110/110 cum laude**
- Weighted average: **29.069/30**
- Certification

2013 – 2018

Scientific high school, applied sciences option, *Liceo A. Rosmini, Rovereto (TN), Italy.*

- Final grade: **100/100**

Experience

2017 – Present

Private Tutor (math and physics)

Private lessons in mathematics and physics for high school and junior high school students. Through this experience I enhanced my communication skills and the ability to explain mathematical/physical concepts to others.

Oct 2020 – Jun 2022

Alba CubeSat Unipd, Mission Analysis team, *University of Padova (IT).*

Alba Cubesat is a team of students who, followed by professors from the University of Padua, aim to design, build and send a CubeSat into orbit, joining the "Fly Your Satellite" Program, promoted by ESA.

June 2017 (2 weeks)

Math and physics laboratory, *University of Trento (IT).*

The activity consisted in participating in seminars with the task of preparing an article in LaTeX related to the topics developed. It was a group work, subject to the revision of a university professor, aimed at deepening the problems related to writing a mathematical text.

Honors & Awards

- **Scholarship "Mille e una lode", A.Y. 2018/2019, 2019/2020, 2020/2021,** *University of Padua (IT).*
  Scholarship for merit awarded to the top 3% students of each degree course.

Relevant additional activities

- National Final of the Mathematics Olympiads in teams (2018).
- Italian Informatics Olympiads in team, regional selections (2017, 2018). Programming language used: C++.
- Second Level Competition of the Italian Physics Olympiads (2017).
- Italian final of the International Mathematical Games established by Bocconi University (years: 2012, 2014, 2016, 2017).

IT skills and proficiency level

Programming Languages	●●●●●	Python, R, LaTeX
	●●●●○	Fortran
	●●●○○	VHDL, Verilog, MatLab
OS and Hardware	●●●●●	Windows, Linux (Ubuntu)
	●●●○○	FPGA
Software, frameworks and libraries	●●●●●	TensorFlow, Keras, Scikit-learn, Git, Docker, Notion
	●●●●○	XGBoost, MySQL
	●●●○○	Vivado, tsfresh, Apache Spark, PySpark, Dask

Projects accomplished

- Koln Traffic Regulator with Parallel Computing
  https://github.com/NicolaZomer/Koln_Traffic_Regulator_with_Parallel_Computing
- Explaining microbial scaling laws using Bayesian inference and Bayesian Data Analysis
  https://github.com/NicolaZomer/Microbial_Scaling_Laws
- Exercises in R of the course Advanced Statistics for Physics Analysis and projects related to the course Statistical Mechanics of Complex Systems
  https://nicolazomer.github.io/R_AdvStatistics_and_ComplexSystems
- Study of the evolution of binary star and binary black holes through the analysis of simulation data
  https://github.com/NicolaZomer/TheFormationOfBinaryBlackHoles
- Sound wave distortion via FPGA using PMOD interface
  https://github.com/NicolaZomer/SoundWaveDistortionViaFPGA
- Exercises of the course Laboratory of Computational Physics
  Module A: https://github.com/NicolaZomer/LaboratoryOfComputationalPhysics_Y4
  Module B: https://github.com/NicolaZomer/ComputationalPhysics_modB
- Assignments done during module A of the course “Management and Analysis of Physics Datasets” (FPGA)
  https://github.com/NicolaZomer/MAPD_modA

Languages

Italian	Mother tongue
English	Full professional proficiency, Cambridge Certificate: B2 (First)
Spanish	Limited working proficiency, not certified

Padova, August 17, 2022

Nicola Zomer